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Exact Minimum-Width Transistor Placement Without Dual Constraint for CMOS Cells

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ABSTRACT

This paper proposes flat and hierarchical approaches for generating a minimum-width transistor placement of CMOS cells in presence of non-dual P and N type transistors. Our approaches are the first exact method which can be applied to CMOS cells with any types of structure. We formulate the transistor placement problem into Boolean Satisfiability (SAT) problem considering the P and N type transistors individually. The experimental results show that our flat approach generates smaller width placement for 29 out of 103 dual cells than that of the conventional method, and the width of only 3 out of 147 cells solved by our hierarchical approach are larger than that of the flat approach. Using the hierarchical approach, 81 % of 340 cells in an industrial standard-cell library of 90 nm technology are solved within one hour for each cell, whereas 32 % using the conventional exact method.

Categories and Subject Descriptors

B.7.2 [**Hardware, Integrated Circuits, Design Aids**]: Layout

General Terms

Algorithms, Design

Keywords

exact minimum-width transistor placement, Boolean Satisfiability, non-dual

1. INTRODUCTION

Cell-based design is one of the major methods for designing VLSI. Automation of standard-cell layout synthesis drastically reduces the required time for creating new standard-cell libraries and can generate high-quality layouts. A lot of papers have been published in the area of the standard-cell layout synthesis. Recently, Guruswamy et al.[1] proposed an automatic P&R method for transistor-level layout generation using Simulated Annealing (SA). This method automatically synthesizes better quality layouts than manually designed layouts. These techniques help the development of the widely used tools in automated standard-cell layout synthesis[2][3].

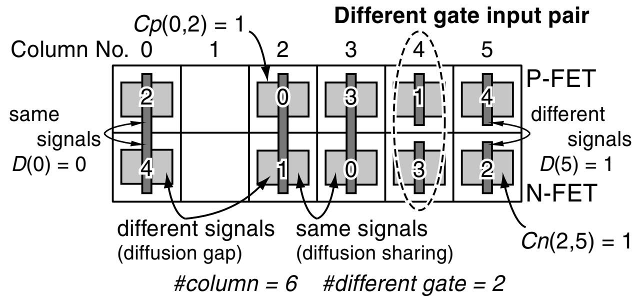

Figure 1: Problem definition of the transistor placement

Several exact cell layout synthesis methods have also been proposed[4][5], since the general style for one dimensional width minimization of static-dual CMOS logic cells was first proposed by Uehara and vanCleemput[6]. However, all these methods make pairs of complementary P and N type transistors and align them in minimum width. Therefore, all these methods can not be applied to some cells that have non-dual P and N type transistors. Moreover, there are some cases that the pairs decided by other rules generate smaller width placements. We have proposed a cell layout synthesis method using Boolean Satisfiability (SAT) previously[7]. This method generates an exact minimum-width transistor placement for non-series-parallel networks, in which there are more than two transistors with a common gate input signal.However, this method is not applicable to the cells including the P and N type transistors which can not be paired by the common gate input signal.

This paper proposes a minimum-width transistor placement method for CMOS cells in presence of non-dual P and N type transistors using SAT. There are two main contributions of this work. First, an exact transistor placement problem of non-dual CMOS cells is defined for the first time. Not only flip flops or tri-state buffers but also intrinsically dual CMOS circuits possibly have non-dual P and N type transistors as a result of transistor folding. As a practical matter, non-dual CMOS cells occupy about a half of an industrial standard-cell library, and all these cells can not be solved by the conventional exact transistor placement methods. Second, the hierarchical approach of the transistor placement is also presented for practical use. Although this approach has possibility to generate wider placement than that generated flatly, the experimental results show that it generates the transistor placements with a little or no increase in cell width. It generates a minimum-width transistor placement hierarchically in about one second even for the cells with large number of transistors, whereas the runtimes of our flat approach is over one minute for the cells with up to 20 transistors and over one hour for larger cells.

Table 1: The variables used for the formulation

name	number	condition which makes the value 1
$C_n(i,k)$	$M \times W$	N-FET i is placed on the column k.
$C_p(i,k)$	$N \times W$	P-FET i is placed on the column k.
$F_n(i)$	M	N-FET i is flipped.
$F_p(i)$	N	P-FET i is flipped.
$D(k)$	W	Different gate input pair is placed on the column k.

2. PROBLEM DEFINITION

When M N type transistors and N P type transistors are given, we have to place these $M+N$ transistors in the minimum area. This problem can be transformed into the problem that places all transistors using the minimum number of columns as illustrated in Figure 1. The P type transistors are aligned in the upper row and the N type transistors are in the bottom. Two neighboring transistors face the diffusions that belong to the common signals each other to connect their source or drain by diffusion sharing. The empty columns result in the diffusion gaps in the final layout. The P and N type transistors placed on the same column form a transistor pair. When the P and N type transistors which make a pair have different gate input signals, this pair is called a "different gate input pair". For example, P type transistor 1 and N type transistor 3 in Figure 1 form a different gate input pair. The number of different gate input pairs is minimized during width minimization of the transistor placement for ease of intra-cell routing which follows the transistor placement step. If a column has only a P or N type transistor, this transistor is also called a different gate input pair. Under these layout style, the transistor placement problem is transformed into the Conjunctive Normal Form (CNF) and Pseudo-Boolean (PB) form constraints. PB form is expressed by the linear inequalities of Boolean variables. We will explain two approaches for generating a minimum-width transistor placement in following sections. First, the flat approach which generates an exact minimum-width placement of any types of CMOS cells will be explained. Next, we will explain the hierarchical approach which reduces the runtime drastically with a little or no increase in cell width.

3. FLAT APPROACH

All variables needed for this formulation are listed in Table 1 where W is the number of the columns of the placement area. Table 1 shows the names of the variable type, the numbers of each type of variables, and the conditions when each type of variables take the value of 1. Some example cases of such conditions are also illustrated in Figure 1. We formulate the following Boolean constraints which express all the possible transistor placements in W columns under our layout style.

Objective function: The number of the different gate input pairs is minimized for ease of intra-cell routing which follows the transistor placement. The objective function is expressed as follows.

$$\text{Minimize}: \quad \sum_{k=0}^{W-1} D(k) \tag{1}$$

Transistor overlap constraint: N type transistors must not overlap in the same column. This constraint is expressed by the following PB constraints.

$$\sum_{i=0}^{M-1} C_n(i,k) \leq 1, \quad 0 \leq k < W \tag{2}$$

The same constraint is expressed as PB constraints for P type transistors.

Transistor appearance constraint: Each N type transistor must be placed on one column. The following PB constraint expresses this constraint.

$$\sum_{k=0}^{W-1} C_n(i,k) = 1, \quad 0 \leq i < M \tag{3}$$

The same constraint is expressed as PB constraints for P type transistors.

Different gate constraint: When a different gate input pair is placed on the column k, the value of $D(k)$ must be 1. This is expressed by the following logic equation,

$$D(k) = \bigvee_{i,j} (C_n(i,k) \wedge C_p(j,k)), \quad 0 \leq k < W \tag{4}$$

$$GATE_n(i) \neq GATE_p(j), \ \ 0 \leq i < M, \ \ 0 \leq j < N \tag{5}$$

where $GATE_n(i)$ and $GATE_p(j)$ mean the gate input signals of N type transistor i and P type transistor j, respectively. This logic equation is transformed into the PB constraints as follows for all combinations of i and j expressed by (5).

$$C_n(i,k) + C_p(j,k) \leq D(k) + 1, \quad 0 \leq k < W \tag{6}$$

$$\text{Minimize}: \quad D(k) \tag{7}$$

Here, the minimization constraint of each $D(k)$ value (7) is already expressed by the objective function (1). Therefore, this constraint is simply expressed by the PB constraints (6) for all combinations of i and j expressed by (5). When the column k has only a P or an N type transistor, $D(k)$ also takes a value of 1. This constraint and the objective function explained before are the keys to handling the CMOS cells including non-dual P and N type transistors.

Neighboring transistors constraint: Two transistors facing the diffusions which belong to the different signals each other can not be placed on the neighboring columns. For example, N type transistors 4 and 1 in Figure 1 can not be placed on the neighboring columns. This constraint is expressed by the following logic equation.

$$GAP_n(i,j) \wedge \bigvee_{k=0}^{W-2} (C_n(i,k) \wedge C_n(j,k+1)) = 0 \tag{8}$$
$$i \neq j, \quad 0 \leq i < M, \quad 0 \leq j < M$$

Here, $GAP_n(i,j)$ is the logic function of $F_n(i)$ and $F_n(j)$, and takes the value of 1 if N type transistor i can not share its diffusion with N type transistor j placed to its immediate right, otherwise 0. This logic equation is expressed as CNF. The same constraint is also expressed for P type transistors.

Finally, we can determine whether given transistors can be placed in W columns or not using these constraints. If no satisfiable assignment is found by the Boolean solver, it is guaranteed that there is no possible placement of W columns. Therefore, we can find an exact minimum-width transistor placement using the procedure described below.

1. For a given transistor netlist, count the number of N and P type transistors. The initial number of column W is set to the same number of *max*(#N-FET, #P-FET).

2. Find a satisfiable assignment for the Boolean constraints constructed for W columns. If a satisfiable assignment is found, these transistors can be placed on the W columns and this procedure terminates. Otherwise, go to step 3.

3. $W = W + 1$. Go to step 2 again.

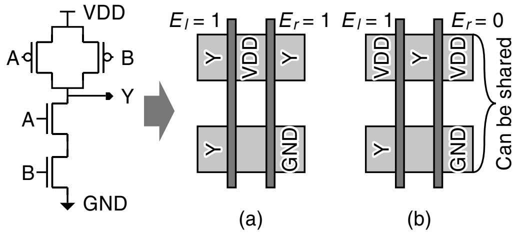

Figure 2: Additional cost introduced to maximize the number of the connections by diffusion sharing between blocks

4. HIERARCHICAL APPROACH

Our hierarchical approach first partitions the given cell into logic blocks. A logic block consists of transistors that are connected together by their diffusions except power and ground. We also recognize a transmission gate as a logic block. It is empirically known that this partitioning procedure does not increase the cell width of the generated placement so much.

Next, intra-block placement of each logic block is performed using the procedure explained in section 3. In this stage, an exact minimum-width transistor placement of each block is generated. When Pseudo-Boolean constraints are generated, one new optimization cost is introduced to maximize the number of the connections by diffusion sharing between logic blocks in inter-block placement step. The diffusions which can be shared between logic blocks are power, ground, and the diffusions of the transmission gates. Therefore it is better to place the diffusions that belong to such signals on the edge of the placement of each block as illustrated in Figure 2 for maximizing the number of the connections by diffusion sharing in the inter-block placement step. We introduce new variables E_l and E_r, and a new objective function into the formulation of transistor placement to maximize the number of the connection by diffusion sharing. E_l (E_r) takes the value of 1 if the diffusions of the left (right) edge do not belong to a pair of the signals which can be shared with other blocks as illustrated in Figure 2. The new objective function is expressed as follows.

$$\text{Minimize}: \quad (W+1) \times (E_l + E_r) + \sum_{k=0}^{W-1} D(k) \tag{9}$$

In our formulation, the number of the connections by diffusion sharing between logic blocks has the higher priority than the number of the different gate input pairs, and (E_l+E_r) have the coefficient $W + 1$ to have larger cost than the different gate input pairs.

Inter-block placement is also based on Boolean Satisfiablility. The constraints explained in section 3 except "different gate constraints" and the "objective function" are needed for the formulation of the inter-block placement. In the formulation of the inter-block placement, each logic block is placed on one column and two logic blocks must not overlap in the same column. Two logic blocks facing the diffusions that can not be shared each other must not be placed on the neighboring column. These constraints are expressed as CNF and PB form constraints in the same manner as explained in section 3. Therefore, we obtain an exact minimum-width block placement using the same procedure as explained in section 3.

Although this approach has possibility to generate wider placement than that of the flat approach, the experimental results show that it generates the transistor placements with a little or no increase in cell width. It generates a minimum-width transistor placement hierarchically in about one second even for the cells with large number of transistors.

Table 2: Cells used for the experiment

Cell number	*Circuit*	*Number of transistors*
1	Series-parallel circuit for $Y = (A \vee B) \wedge C$	10
2	Tri-state buffer	12
3	2-input multiplexer	14
4	Series-parallel circuit for $Y = \overline{(A \wedge B) \vee (C \wedge D) \vee (E \wedge F) \vee (G \wedge H)}$	16
5	Series-parallel circuit for $Y = (A \wedge B) \oplus C$	18
6	3-input exclusive OR	20
7	3-input exclusive NOR	22
8	D Flip Flop	24
9	D Flip Flop (buffered)	26
10	3-input exclusive NOR (buffered)	28

5. EXPERIMENTAL RESULTS

5.1 Flat Approach

The proposed flat and hierarchical transistor placement methods were implemented and the list of the CNF and PB form constraints explained in sections 3 and 4 were automatically created. In this experiment, we used PBS[8] for a Boolean constraint solver. PBS can handle CNF constraints, PB form constraints, and optimizations. The characteristics of the cells used in this experiment are shown in Table 2. Table 3 shows the comparison with the conventional exact method[7] which is based on SAT. The conventional method is applicable to the cells that can not be applied to other exact methods such as [4] and [5]. Moreover, the conventional method theoretically generates the same or smaller width placement than that of other existing exact transistor placement methods.

Table 3 compares the problem sizes of each formulation, the numbers of columns of generated placement, and the total runtimes of each method for generating the minimum-width transistor placement from a netlist. In the column of problem size, "#variable", "#clause", and "#inequality" mean the number of variables, clauses, and inequalities of each formulation, respectively. In the column of width, "#column" and "#different" mean the number of columns and the number of the different gate input pairs of generated placement, respectively. The proposed method used PBS[8] for a Boolean constraint solver, and the conventional method used Chaff[9] for a CNF-SAT solver. The conventional method can not be applied to the cells such as a tri-state buffer (cell number 2), because they have transistors which can not be paired by the common gate input signal. In the case of the cell number 9, the runtime was over 3600 seconds for the proposed method, and that of both methods were over 3600 seconds in the case of the cell number 10. In the other cases, the proposed method generated the same or smaller width placements by introducing different gate input pairs. Although the intra-cell routing may become difficult by introducing the different gate input pairs, it is easy to complete it if using the second metal layer.

5.2 Hierarchical Approach

Table 4 shows the comparison result between our flat and hierarchical approaches. This table compares the numbers of the columns and the different gate input pairs of the generated placements, and the total runtimes. The width of the generated placement increased in the case of the cell number 1, and the numbers of different gate input pairs increased in the cases of the cells number 5, 6, and 7. This table clearly shows that the runtimes for transistor placement was drastically reduced by our hierarchical approach with a little or no increase in cell width. Moreover, this hierarchical approach could solve the cells with larger number of transistors. The solu-

Table 3: Detailed comparison results with the conventional exact method

Cell number	Problem Size					Width			Runtime (sec.)	
	Conventional		Proposed			Conv.	Proposed		Conv.	Proposed
	#variable	#clause	#variable	#inequality	#clause	#column	#column	#different		
1	40	760	65	121	488	6	**5**	2	0.03	0.04
2	N/A	N/A	103	251	1104	N/A	7	2	N/A	0.15
3	70	2522	103	251	1104	9	**7**	2	0.86	0.17
4	80	2928	152	497	2632	9	**8**	2	0.27	0.45
5	90	4588	208	699	4212	10	10	0	1.66	2.03
6	100	6412	251	1011	6320	14	**11**	2	55.82	29.06
7	110	8478	321	1427	9120	15	**13**	2	159.63	969.96
8	144	17122	349	1663	11040	18	**13**	4	1382.98	1240.83
9	130	13070	404	2127	14170	*	*	*	>3600	>3600
10	140	15516	434	2465	15860	16	*	*	1730.13	>3600

* means that the procedure does not terminate after 3600 seconds

Table 4: Comparison results between our flat and hierarchical approaches

Cell number	Width				Runtime (sec.)	
	Flat		Hier.		Flat	Hier.
	#col.	#diff.	#col.	#diff.		
1	5	2	**6**	0	0.04	0.02
2	7	2	7	2	0.15	0.02
3	7	2	7	2	0.17	0.14
4	8	2	8	2	0.45	0.45
5	10	0	10	**2**	2.03	0.08
6	11	2	11	**4**	29.06	0.29
7	13	2	13	**4**	969.96	3.39
8	13	4	13	4	1240.83	0.31
9	*	*	15	4	>3600	1.21
10	*	*	16	4	>3600	1.85

* means that the procedure does not terminate after 3600 seconds

Table 5: Comparison of the number of cells that can be applied to the proposed exact method and the conventional one

placement method	applicable		solvable	
	#cells	coverage	#cells	coverage
Conventional	164	48 %	110	32 %
Proposed (flat)	340	100 %	147	43 %
Proposed (hier.)	340	100 %	274	81 %

tions for the cells number 9 and 10 were generated in less than two seconds by the hierarchical approach, whereas the flat approach could not generate the solutions in over one hour.

Table 5 compares the numbers of cells that can be applied to the conventional exact method[7] and the proposed flat and hierarchical methods in the case of an industrial standard-cell library of 90 nm technology including 340 cells. This table shows the numbers of the cells and coverage ratios of the cells that are applicable to each method and that can be solved in 3600 seconds. Because the proposed method is applicable to CMOS cells with any types of structure, this method was applied to all of the 340 cells, whereas the conventional one was about a half. The coverage ratios of the cells that can be solved in 3600 seconds by the flat approach and the conventional method are 43 % and 32 %, respectively. Although the number of cells which can not be solved by the proposed flat approach was larger than that of the conventional one, the coverage ratio of the solvable cells of the flat approach was larger than that of the conventional one because the proposed method can be applied to all the cells. Among the cells solved by both methods, the proposed flat approach generated smaller width placements for 29 out of 103 dual cells by introducing the different gate input pairs. Using the hierarchical approach, the coverage ratio increased to 81 %, and only 3 out of 147 cells have wider placement than the placement generated by the flat approach by one column. This result shows that our hierarchical approach reduced the runtimes drastically with a little or no increase in cell width.

6. CONCLUSIONS

This paper proposed flat and hierarchical approaches for generating a minimum-width transistor placement of CMOS cells in presence of non-dual P and N type transistors. Our approaches are the first exact minimum-width transistor placement method for non-dual CMOS cells. The experimental results showed that it is not only applicable to CMOS cells with any types of structure, but also effective even for dual CMOS cells. The hierarchical approach which is based on circuit partitioning reduced the runtimes drastically at the cost of increased area in few cases. These results show that our approaches can be used for building high-quality standard-cell libraries.

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